

Microsoft Sentinel Hunting Guide

Project 3 – Brute Force & Account Compromise Investigation

Purpose

This guide documents the Microsoft Sentinel hunting methodology used to investigate a simulated brute force attack, successful account compromise, privilege escalation, and post-compromise discovery activity in an Azure-based SOC lab.

Environment

Azure VM (Windows 11 Endpoint), Microsoft Defender for Endpoint, Microsoft Sentinel, Splunk Enterprise, Sysmon, Windows Security Logs.

Primary Data Sources

SecurityEvent, DeviceProcessEvents, DeviceLogonEvents, DeviceNetworkEvents, DeviceRegistryEvents, DeviceEvents.

Investigation Workflow

1. Validate alerts in Sentinel Incidents.
2. Pivot into the affected endpoint.
3. Review device timeline.
4. Correlate successful (4624) and failed (4625) logons.
5. Review PowerShell execution.
6. Review account privilege changes.
7. Review discovery commands (hostname, whoami, ipconfig, netstat, tasklist).
8. Correlate evidence with Defender timeline and Splunk.

Recommended KQL Hunts

Failed logons:

SecurityEvent | where EventID==4625

Successful logons:

SecurityEvent | where EventID==4624

PowerShell:

DeviceProcessEvents

| where FileName =~ "powershell.exe"

Privilege enumeration:

DeviceProcessEvents

| where ProcessCommandLine has_any ("whoami /priv", "whoami /groups")

Network discovery:

DeviceProcessEvents

| where ProcessCommandLine has_any ("ipconfig", "netstat", "hostname", "tasklist")

Indicators of Compromise

Repeated Event ID 4625, Event ID 4624 following failures, PowerShell execution, discovery commands, administrator group enumeration, suspicious network reconnaissance.

Evidence Collected

Windows Event Viewer, Microsoft Defender Device Timeline, Microsoft Sentinel hunting queries, Splunk dashboards, Sysmon process creation logs.

MITRE ATT&CK; Mapping

T1110 Brute Force; T1078 Valid Accounts; T1059.001 PowerShell; T1057 Process Discovery; T1082 System Information Discovery; T1033 Account Discovery; T1069 Permission Groups Discovery; T1016 Network Configuration Discovery; T1049 Network Connections Discovery.

Analyst Conclusions

The investigation reconstructed the complete attack lifecycle from brute-force authentication attempts through successful logon and post-compromise discovery. Cross-platform telemetry demonstrated effective detection and correlation across Sentinel, Defender, Splunk, and Sysmon.